

PUSHPAK PATIL

AWS Cloud Architect | DevOps Engineer | FinOps & AIOps Product Builder

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AWS Certified Solutions Architect – Associate | AWS Certified Security – Specialty | IBM Vault Automation Practitioner

PROFESSIONAL SUMMARY

Dynamic and results-driven Cloud Solution Architect and DevOps Engineer with 4+ years of progressive experience designing, developing, and implementing large-scale, multi-cloud infrastructure solutions for multinational financial institutions. Expert in AWS architecture, Python backend development, IoT-enabled cloud services, and real-time data processing. Proven expertise in Infrastructure as Code (Terraform, CloudFormation), Kubernetes/EKS orchestration, CI/CD pipeline automation, and full software development lifecycle management. Track record of achieving up to 60% cost reduction, 100% compliance scores (PCI DSS, ISO 27001), and implementing ML-powered monitoring and anomaly detection platforms. Experienced in cloud migrations, disaster recovery design, time-series data pipelines, and leading cross-functional teams in regulated environments.

CORE COMPETENCIES & TECHNICAL SKILLS

Cloud Platforms	AWS (4+ Years): EC2, EKS, ECS, Lambda, S3, RDS, DynamoDB, IoT Core, IoT Analytics, Kinesis, SageMaker, Control Tower, Organizations, Security Hub, GuardDuty, Config. Microsoft Azure: Azure VMs, Networking, AD, DevOps, Cross-Cloud Migration.
Architecture & Security	High-Availability & Disaster Recovery, Multi-AZ/Multi-Region Design, Cloud Security Posture Management (CSPM), IAM, WAF & Shield, PCI DSS, CIS Benchmarks, ISO 27001, Security Audits & Compliance Reviews, Zero-Trust Architecture.
Backend Development	Python (Flask, FastAPI, Django), Node.js, RESTful API design, microservices architecture, serverless backend (Lambda), event-driven architectures, Golang exposure.
IoT & Real-Time Data	AWS IoT Core, IoT device provisioning, MQTT protocol, real-time sensor data ingestion, time-series data processing, streaming analytics, Kinesis Data Streams/Analytics, IoT-enabled cloud service infrastructure.
ML & Data Pipelines	ML model deployment (SageMaker, Lambda), time-series forecasting, anomaly detection algorithms, ETL processes, data pipeline orchestration, real-time ML inference.
Infrastructure as Code	Terraform, Terragrunt, AWS CloudFormation, Python automation, Shell Scripting, Lambda State Machines, Step Functions, Policy-as-Code.
DevOps & CI/CD	Jenkins, GitHub Actions, GitLab CI, automated deployment pipelines, blue/green & canary deployments, Docker containerization, software development lifecycle, agile practices.
Monitoring & Observability	Prometheus, Grafana, PromQL, CloudWatch, OpenTelemetry, AIOps platforms, ML-based anomaly detection, automated alerting, log aggregation.
Networking	VPC Design, Transit Gateway, Direct Connect, Hybrid Connectivity (VPN/MPLS), Security Groups, NACLs, Route 53, Load Balancers, Palo Alto Firewalls, AWS Global Accelerator.
Databases & Migration	RDS (MySQL, PostgreSQL, Oracle), DynamoDB, DMS, Database Replication & Optimisation, Cross-Cloud Migration, time-series databases.

PROFESSIONAL EXPERIENCE

Cloud Solution Architect | Applied Cloud Computing, Thane, India | March 2025 – Present

- Architect and implement scalable AWS cloud infrastructure supporting IoT-enabled services, real-time sensor data ingestion, ML model deployment, and backend API services for multinational financial technology clients
- Design and develop Python backend services using Flask and FastAPI, implementing RESTful APIs for IoT device management, time-series data processing, and ML inference endpoints
- Manage complete software development lifecycle from requirements gathering through deployment, implementing agile methodologies, DevOps best practices, and CI/CD automation for continuous delivery
- Build and maintain data pipelines for real-time sensor data processing using AWS Kinesis, Lambda, and DynamoDB, enabling time-series analytics and ML-powered operational insights
- Lead DevOps automation: provisioning cloud infrastructure with Terraform, configuring monitoring stacks (Prometheus, Grafana), and implementing automated deployment pipelines

Senior Cloud Engineer | Applied Cloud Computing, Mumbai, India | March 2022 – February 2025

- Designed and implemented AWS Landing Zones with Terraform for financial institutions (SBGIC, Zuno, HDBFS), managing complex multi-account environments with Control Tower
- Developed Python automation scripts for cloud provisioning, compliance monitoring, cost optimization, and operational workflow automation across client portfolios
- Implemented DevOps pipelines using Jenkins and GitLab CI for automated testing, deployment, and infrastructure updates across development, staging, and production environments
- Conducted comprehensive Security Audits and Compliance Reviews (PCI DSS, CIS), achieving 100% compliance scores through automated monitoring and remediation
- Drove cost-saving initiatives achieving 20-60% expense reduction through resource rightsizing, serverless adoption, and optimised billing strategies
- Provided 24x7 support for banking clients (IBDIC, Bandhan Bank), troubleshooting complex infrastructure issues and cloud incident resolution
- Mentored junior cloud engineers and conducted internal AWS training sessions to upskill team members

Founder & Tutor | Elite Educare Academy, Thane, India | March 2015 – March 2022 (7 years)

- Founded and managed educational institution with a team of 20 teachers across Thane, teaching mathematics, physics, cloud services, security, Linux administration, and AWS
- Trained 350+ students, developing strong communication, team leadership, and the ability to simplify complex technical concepts
- Skills: Cloud Computing, Linux Administration, AWS, Networking, Security Fundamentals, Team Leadership, Python Programming

KEY PROJECTS & ACHIEVEMENTS

Cloud Solution Architect Projects (Architecture, Migration, IoT & AIOps)

IoT Banking Platform	IoT-Enabled Cloud Services & Real-Time Data Processing Architected scalable AWS IoT infrastructure for real-time ATM and branch sensor monitoring. Built Python backend services for IoT device provisioning, data ingestion APIs, and time-series anomaly detection using ML models deployed on SageMaker. Achievement: Reduced ATM downtime by 45%, enabled predictive maintenance, sub-second latency for real-time alerts, processing 1M+ sensor readings/day.
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Bandhan Bank	AIOps, Security Automation & ML-Based Analytics Implemented AIOps platform with ML-based anomaly detection for time-series metrics, AI-powered root cause analysis, and auto-remediation using AI agents and Lambda workflows. Built Python data pipelines for metrics ingestion and ML model training. Achievement: Reduced incident response from hours to minutes, 100% audit readiness, zero critical findings in external security audits.
ATLAS	AI-Powered Automated DR Orchestration with Multi-Region Resilience Designed fully automated Disaster Recovery solution with AI-driven health monitoring, intelligent multi-region failover/failback using Lambda State Machines and Python orchestration layer. Achievement: Zero-touch failover with RTO under 10 minutes, reduced DR testing from 8 hours to 30 minutes.
RBL Bank	Cross-Cloud Migration (Azure to AWS) Led migration of GitHub Enterprise Server from Azure to AWS using Terraform, including version upgrade and optimal AWS infrastructure design. Achievement: Zero data loss, 45% performance improvement, 30% cost reduction vs. Azure.
APAC	Multi-Region DR & Database Migration Architected cross-region DR with automated replication and migrated Oracle/MySQL to RDS Multi-AZ. Achievement: RTO 15 min, RPO 5 min, 100% DR drill success rate, 60% reduction in DB admin overhead.
TMF	EKS Performance Optimisation & Kubernetes Troubleshooting Diagnosed and resolved critical memory leaks through container profiling and fixed API throttling. Implemented HPA/VPA with optimised resource limits. Achievement: 98% reduction in OOMKilled restarts, deployment success 60% to 99.5%, cluster uptime 99.95%.

Senior Cloud Engineer Projects (Implementation, Compliance & Cost Optimisation)

SBIGIC	Enterprise Cloud Migration & Network Design Led end-to-end AWS migration for insurance workloads. Designed multi-AZ VPC with Direct Connect/VPN hybrid connectivity and layered security. Achievement: 100+ servers migrated, 99.95% uptime, production stability in 2 weeks, MTTR reduced 35%.
HDBFS	Compliance Automation & Cost Optimisation with Terraform Built automated compliance monitoring using Config Rules, Security Hub, and Terraform. Implemented cost optimisation framework with Lambda/Step Functions. Achievement: 100% compliance score, 60% cost reduction, automated reporting saving 95% manual effort.
Edelweiss	Microservices Migration to EKS & DevOps Modernisation Rearchitected 50+ monolithic apps to cloud-native microservices on EKS/Fargate with Terraform, CI/CD pipelines, and Transit Gateway connectivity. Achievement: 35% cost reduction, deployment frequency monthly to daily, enhanced scalability.

CERTIFICATIONS & EDUCATION

Professional Certifications

- AWS Certified Solutions Architect – Associate
- AWS Certified Security – Specialty
- IBM Vault Automation Practitioner Advanced

Education

- Master of Science in Organic Chemistry – Mumbai University (2015-2017)
- Bachelor of Science in Mathematics & Computer Science – Mumbai University (2014-2015)

PRODUCTS BUILT

Independently designed, developed, and commercialised cloud-native SaaS products for the BFSI sector:

Nimbus FinOps AI	Multi-Tenant Cloud FinOps Platform for BFSI finops.acc.ltd/dashboard github.com/p05-cloud/zuno-finops Built a production-grade, multi-tenant Cloud FinOps SaaS platform for banking and financial services clients. Deployed live at Poonawalla Fincorp (PFL), BHFL, and Axis Securities. Integrates AWS, Azure, GCP, and Kubernetes billing under a single FOCUS-spec normalised data model with AI-powered cost recommendations, anomaly detection, rightsizing engine, savings plan optimisation, tagging governance, and budget alerting. Tech Stack: Next.js 15 (App Router, Server Components), TypeScript, PostgreSQL 16, Redis 7, BullMQ (background ingestion workers), Prisma ORM, Zod validation, Tailwind CSS, shadcn/ui, Docker, Helm (Kubernetes), GitHub Actions CI/CD, Vercel. Architecture: Monorepo (pnpm workspaces + Turborepo) with FOCUS schema normalisation, multi-tenant Row-Level Security in PostgreSQL, BullMQ ingestion workers consuming AWS CUR, Azure Cost Management, and GCP BigQuery billing exports. Supports Docker Compose, Kubernetes/Helm, Vercel, and bare-metal deployments. Achievement: Live in production at 3 BFSI clients. Reduced manual cost review effort by 80%, automated compliance governance across 4+ cloud accounts per client.
InfraOps AI	AIOps Platform for Automated L1 Incident Remediation github.com/p05-cloud/infraops-ai Building an AI-powered AIOps platform designed to eliminate L1 operational errors through intelligent monitoring, pattern recognition, and automated remediation workflows. Uses ML models to detect infrastructure anomalies, classify incident severity, and trigger auto-remediation scripts via AWS Systems Manager and Lambda — reducing on-call burden and MTTR for cloud operations teams. Tech Stack: AWS CloudWatch, Lambda, Systems Manager, Prometheus, Grafana, Python (ML inference), event-driven architecture. Achievement: In active development. Target: 70% reduction in L1 ticket volume and sub-5-minute automated remediation for common cloud infrastructure failures.

AWARDS & RECOGNITION

- Cross-Team Collaboration Excellence Award
- Excellence in Cloud Cost Management Award
- Rooky Award – Applied Cloud Computing (ACC)
- Persistence Award – Applied Cloud Computing (ACC)
- Anchor Project Excellence Award
- Organisational Growth Leader Award
- Led Disaster Recovery (DR) drills and restoration testing across client accounts

- Created 18 cloud infrastructure automations for process optimization and time savings